

Kadhirvel M

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Objective

Research and Engineering Scientist with expertise in **independent research, system design, and emerging technologies**. Skilled in building autonomous AI systems, orchestrating multi-agent pipelines, and deploying scalable **LLM** and generative AI solutions in **cloud computing** and production environments. Experienced in **technical documentation, drafting specifications, and collaborative product development**, seeking a **Research and Engineering Scientist** role at **Zoho** to contribute to innovative AI, computing, and system design research. **Additionally, proficient in compiler design and its application in AI systems.**

Education

Manakula Vinayagar Institute of Technology 🔗 <i>BTech in Artificial Intelligence & Machine Learning</i>	<i>Sep 2022 – May 2026</i> CGPA: 8.42/10
◦ Coursework: Deep Learning, NLP, Computer Vision, Transformers, Multimodal Learning, AI Ethics, Algorithms, Computer Architecture, Data Structures, Software Engineering, and Computer Science.	

Experience

Agentic AI Research Intern <i>Tuta Consultancy Services (TCS)</i> 🔗	<i>Chennai, India</i> <i>June 2025 – Aug 2025</i>
◦ Conducted independent research and novelty identification in multi-agent AI systems for automated package migration across cloud platforms.	
◦ Designed and implemented system architecture, technical specifications, and AI pipelines using Autogen, MCP, Python, Azure, and GCP . Collaborated with support teams to enhance system supportability .	
◦ Deployed LLMs and generative models in production via Docker and Kubernetes , ensuring scalability, reliability, and maintainability.	
◦ Generated and augmented synthetic data, performed data cleaning and parsing , and optimized model inputs for machine learning, NLP, and image processing tasks .	
◦ Drafted detailed technical documentation, reports, and presentations for internal and external communication, including miscellaneous insights on AI topics.	

Technical Developer <i>Ignite Skylabs</i> 🔗	<i>Puducherry, India</i> <i>Mar 2024 – May 2024</i>
◦ Built and deployed full-stack AI applications, improving system efficiency and client satisfaction.	
◦ Conducted team workshops and mentoring in Tamil, emphasizing collaboration, task prioritization, and knowledge sharing in web technologies.	
◦ Produced project documentation and specifications, supporting long-term maintenance and product lifecycle.	

Skills

Computer & Engineering Systems::

- Computer Architecture, Computer Networking, Data Structures, Software Engineering, Programming Languages, Image Processing, Robotics, Automation

AI & Multimodal Modeling:

- Autogen, MCP, Transformers (GPT, BERT, RoBERTa), GANs, VAEs, LLaMA, PyTorch, TensorFlow

Cloud & Containerizations:

- AWS, Azure, GCP, Kubernetes, Docker, Production Environments

NLP & Text Processing:

- Text Cleaning, Parsing, Sentiment Analysis, Text Generation, Chunking, Embeddings

Data Management & Synthetic Data:

- Synthetic Data Generation, Data Labeling, Augmentation, NLP Pipelines, Datasets

AI Pipelines & Deployment:

- Modeling, Cloud Deployment, Optimization, Monitoring, DevOps

Soft Skills:

- Analytical, Innovative, Responsible, Collaborative, Problem-Solving, Independent, Attention to Detail, Organization, Task Prioritization, Communication, Supportive

Publications

IEEE Xplore: WiFi-based Attendance Detection	Dec 2024
Conference: AI-Powered Neurodiagnostics	Apr 2024
IEEE Xplore: Fake News Detection on Social Media	Nov 2023

Projects

Autonomous Package Migrator (TCS)

- Multi-agent GenAI system for codebase translation across stacks, using Autogen, MCP, and Python.
- Developed system design, technical specifications, and deployment architecture for containerized microservices via Kubernetes on Azure.

Alzheimer's & Parkinson's - Neurodegenerative Disease prediction

- Built EfficientNet B7 with autoencoders and attention modules, improving diagnosis accuracy by 40%.
- Deployed models via AWS SageMaker, demonstrating machine vision, image processing, and biomedical AI research.

Agentic RAG Database Chatbot

- Designed a Retrieval-Augmented Generation (RAG) chatbot integrated with a knowledge database for dynamic query responses.
- Implemented using Autogen, MCP, and Python, with multi-agent orchestration for context-aware responses.
- Developed system design, technical specifications, and backend architecture, deployed on cloud infrastructure for scalable production use.

Certifications

Data Analytics - LinkedIn 🔗	<i>Sept, 2021</i>
Foundations: Data, Data, Everywhere - Coursera 🔗	<i>Oct, 2021</i>
Google Project Management - Coursera 🔗	<i>Mar, 2022</i>
Big Data in the Age of AI - LinkedIn 🔗	<i>Sept, 2021</i>
Development & Advanced Engineering - Accenture 🔗	<i>Nov, 2024</i>

Achievements

- **Winner** – *Castilo'25 Tech Hackathon*, March 2025
- **Runner-up** – *SCIMIT'25 Project Expo*, February 2025
- **Runner-up** – *Frontend Fusion'24* by Datanerdz AI (Web Development Competition), July 2024